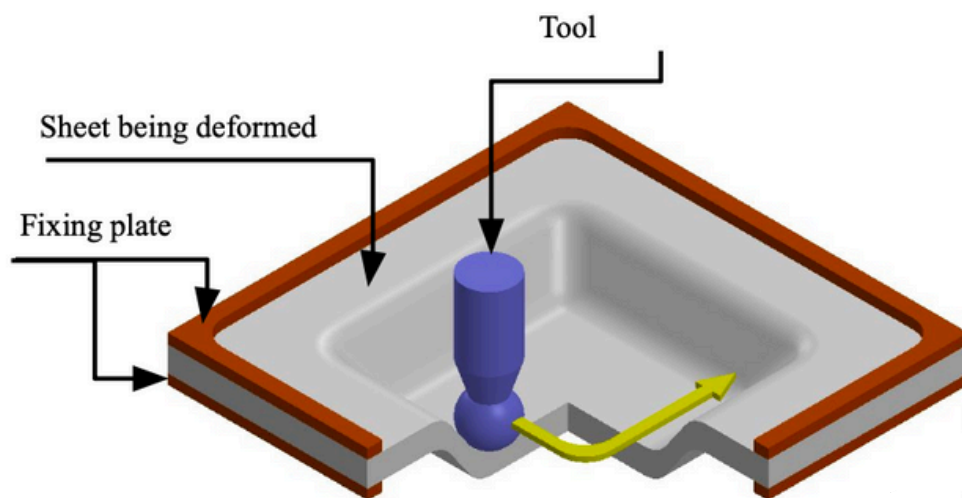


TOOL PATH GENERATION USING ISF SLICER METHOD FOR 3D SOLID SHAPES

WHAT IS INCREMENTAL SHEET FORMING (ISF)?

INCREMENTAL SHEET FORMING (ISF) IS A DIE-LESS SHEET METAL FORMING PROCESS THAT INCREMENTALLY DEFORMS A SHEET USING A CNC-CONTROLLED HEMISPHERICAL TOOL. UNLIKE TRADITIONAL FORMING PROCESSES THAT REQUIRE EXPENSIVE DIES, ISF PROVIDES GREATER FLEXIBILITY, COST-EFFECTIVENESS, AND MASS CUSTOMIZATION CAPABILITIES.



KEY ADVANTAGES OF ISF:

- FLEXIBILITY: NO NEED FOR CUSTOM DIES, MAKING IT IDEAL FOR PROTOTYPING AND LOW-VOLUME PRODUCTION.
- COST REDUCTION: ELIMINATES DIE COSTS, REDUCING MANUFACTURING EXPENSES.
- HIGHER FORMABILITY: CAN ACHIEVE COMPLEX GEOMETRIES WITH HIGHER FORMING LIMITS THAN TRADITIONAL METHODS.
- RAPID PROTOTYPING: ALLOWS QUICK TESTING AND ITERATION OF NEW DESIGNS.

VARIANTS OF ISF WITH ADDITIONAL DETAILS:

1. SINGLE POINT INCREMENTAL FORMING (SPIF):

- THE MOST BASIC FORM, WHERE A SINGLE TOOL DEFORMS THE SHEET METAL WITHOUT A SUPPORT DIE.
- TOOLPATH CAN FOLLOW CONTOUR-BASED OR SPIRAL-BASED TRAJECTORIES.
- SUITABLE FOR RAPID PROTOTYPING AND SMALL-BATCH PRODUCTION.

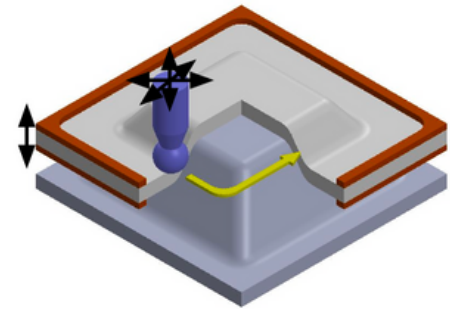
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2. TWO-POINT INCREMENTAL FORMING (TPIF):

- USES A PARTIAL OR FULL DIE SUPPORT TO ENHANCE GEOMETRIC ACCURACY AND REDUCE SHAPE DEVIATION.
- MINIMIZES SPRING-BACK EFFECTS AND IMPROVES FINAL PART QUALITY.



Single Point Incremental Forming (SPIF)



Two Point Incremental Forming (TPIF)

KEY REQUIREMENTS FOR PERFORMING ISF:

• TOOLING REQUIREMENTS

- FORMING TOOL: A HEMISPHERICAL TOOL WITH A SMOOTH SURFACE TO PREVENT SCRATCHING.
- MADE FROM HARDENED STEEL, CARBIDE, OR CERAMIC FOR DURABILITY.

• CNC MILLING MACHINE

- A 3-AXIS CNC MILLING MACHINE IS COMMONLY USED, BUT A 5-AXIS MACHINE IS PREFERRED FOR COMPLEX GEOMETRIES.

• FIXTURES FOR HOLDING THE WORKPIECE

- SIMPLE CLAMPING FRAME
-

• A CAD MODEL (.STEP FILE)

WHAT DOES THE SOFTWARE DO?

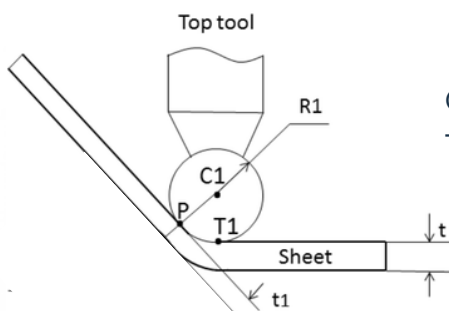
THIS SOFTWARE AUTOMATES TOOLPATH GENERATION FOR ISF BY PROCESSING STEP (.STP) FILES, EXTRACTING GEOMETRIC FEATURES, AND GENERATING TOOLPATHS OPTIMIZED FOR CONTOUR-BASED AND SPIRAL-BASED FORMING STRATEGIES.

PRE-PROCESSING & GEOMETRIC FEATURE EXTRACTION:

- READS STEP FILES USING OPENCASCADE.
- IDENTIFIES PART GEOMETRY AND SLICING HEIGHTS FOR TOOLPATH GENERATION.
- HANDLES COMPLEX SURFACES WITH VARYING FILLETS, EDGES, AND CONTOURS.

TOOLPATH PLANNING WITH COMPENSATION:

- CONTOUR-BASED TOOLPATH: CREATES LAYER-WISE PATHS WHERE EACH CONTOUR FOLLOWS A SPECIFIC DEPTH INCREMENT.
- SPIRAL-BASED TOOLPATH: GENERATES A CONTINUOUS TRAJECTORY THAT AVOIDS TOOL SHIFTING FROM CONTOUR TO CONTOUR ALSO IMPROVES SURFACE QUALITY.
- TOOL COMPENSATION:
 - GENERATES TOOL RADIUS COMPENSATED PATH TAKING TOOL TIP AS THE REFERENCE POINT.



$$C1 = P + R1 \times n$$
$$T1 = C1 - R1 \times k$$

Where-

- P: Contact point between top tool and sheet.
- C1: Center of top tool's roller.
- T1: Bottom contact point of the top tool.
- R1: Radius of the top tool.
- n: Unit normal vector to the sheet at P.

SIMULATION & VISUALIZATION:

- 3D REAL-TIME VISUALIZATION USING PLOTLY TO PREVIEW TOOL MOTION.
- ANIMATION OF TOOLPATHS FOR TRAJECTORY VALIDATION BEFORE CNC EXECUTION.

G-CODE GENERATION & EXPORT:

- GENERATES G-CODE FOR STANDARD CNC MACHINES.
- SUPPORTS COMMON CONTROLLERS SUCH AS FANUC AND SIEMENS.
- OFFERS CUSTOMIZATION OF FEED RATES, TOOL STEP-DOWN, AND SPINDLE SPEED.

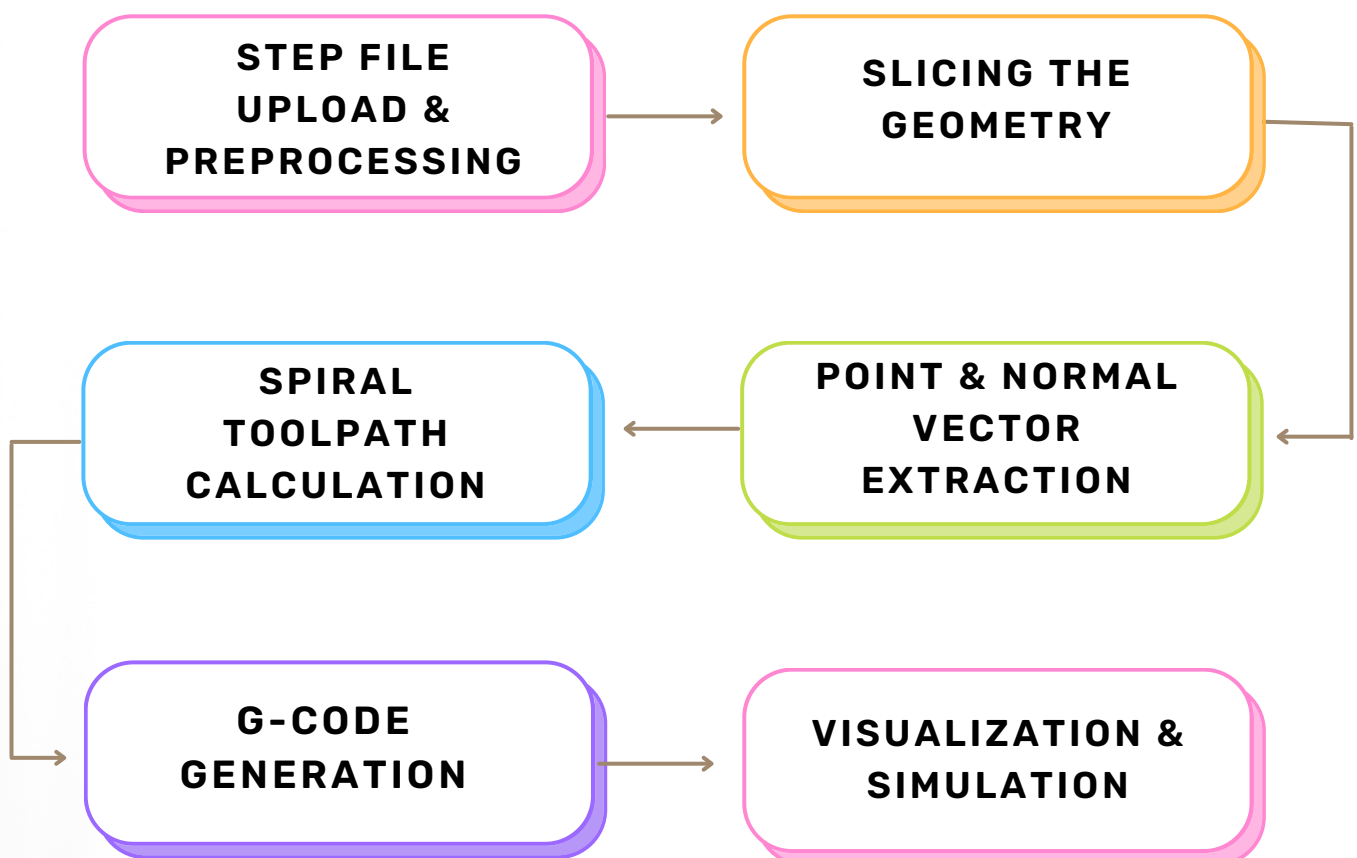
ERROR HANDLING & USER INTERFACE:

- DETECTS GEOMETRICAL INCONSISTENCIES IN THE IMPORTED STEP FILE.
- PROVIDES AN INTUITIVE WEB-BASED USER INTERFACE USING FLASK.
- ALLOWS USERS TO MODIFY PARAMETERS AND RECALCULATE TOOLPATHS DYNAMICALLY.

USER MANAGEMENT & FEEDBACK INTEGRATION:

- SECURE LOGIN AND USER PROFILES FOR STORING PROJECT FILES.
- FEEDBACK MECHANISM FOR USERS TO REPORT ERRORS OR REQUEST IMPROVEMENTS.

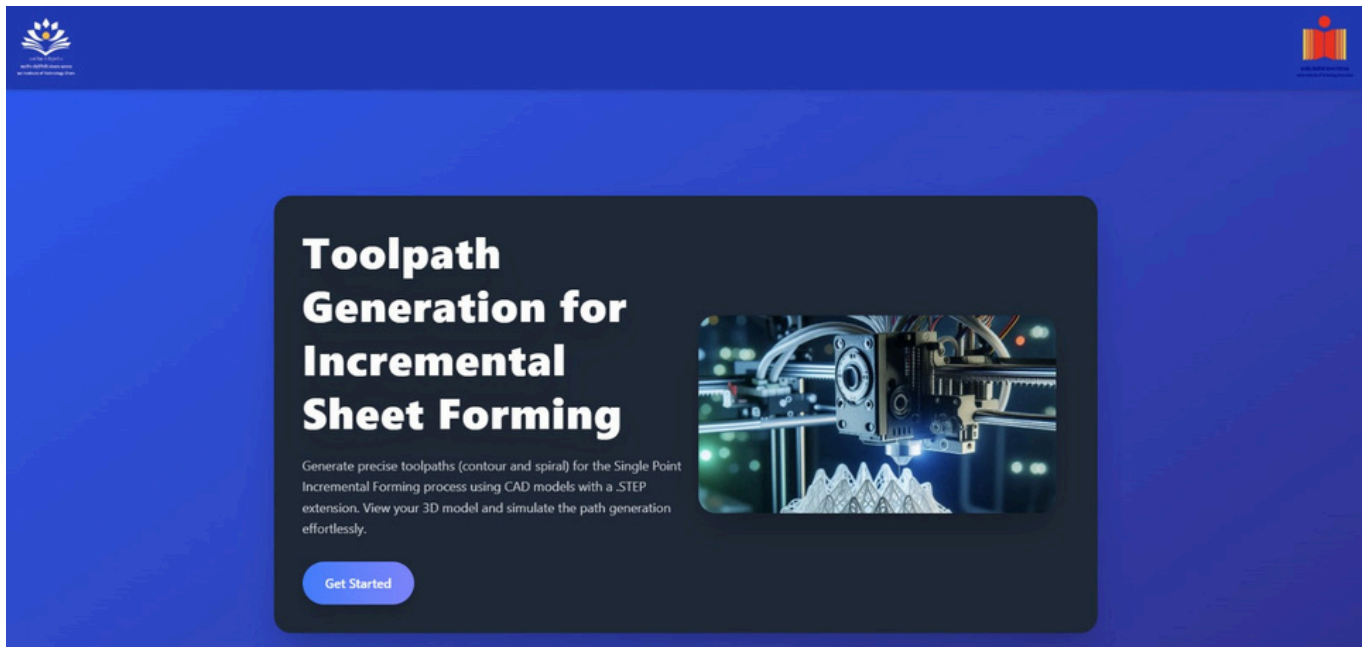
FLOWCHART





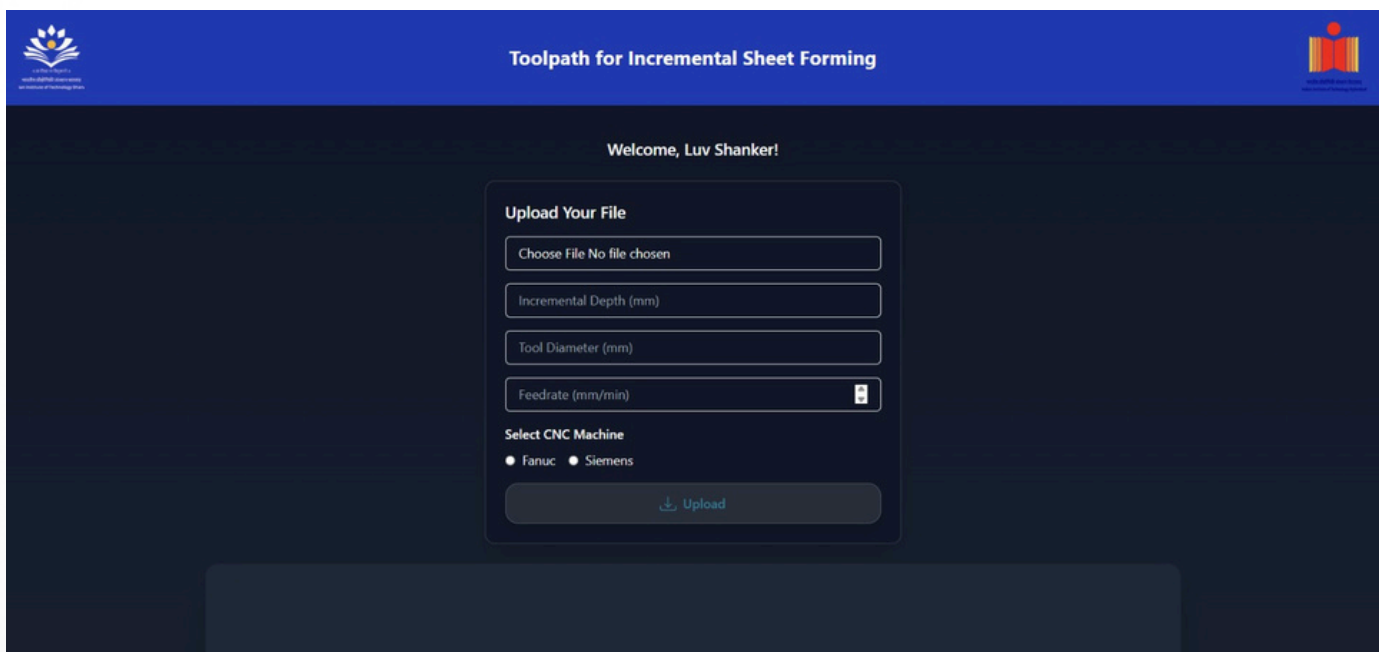
WEBSITE WORKFLOW:

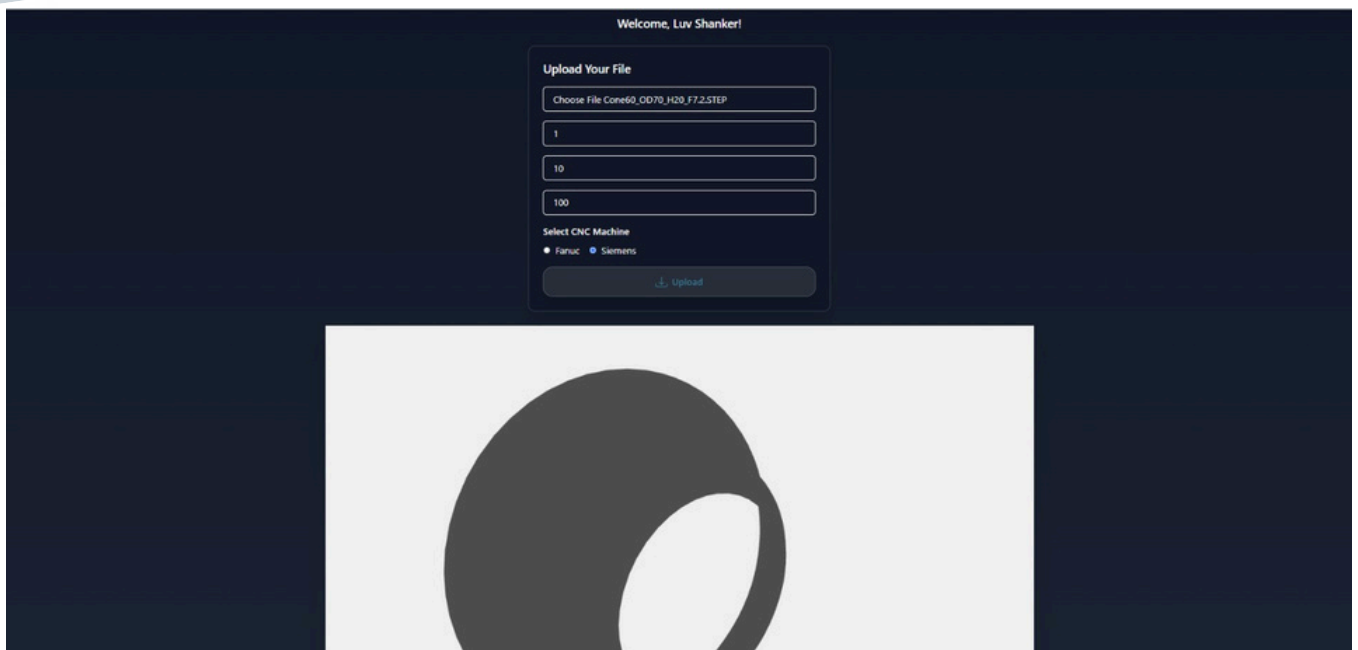
- THIS IS THE LANDING PAGE FOR THE TOOLPATH GENERATION FOR INCREMENTAL SHEET FORMING TOOL. CLICK ON "GET STARTED" TO LOG IN AND BEGIN GENERATING THE TOOLPATH.



- UPLOAD PAGE ENTER ALL REQUIRED INPUTS TO GENERATE THE TOOLPATH. UPLOAD YOUR .STEP FILE, ENTER INCREMENTAL DEPTH, TOOL DIAMETER, FEEDRATE, AND SELECT THE CNC MACHINE TYPE (FANUC OR SIEMENS), THEN CLICK UPLOAD TO PROCEED.

AFTER UPLOADING, THE SELECTED CAD FILE WILL BE AVAILABLE FOR THE USER TO PREVIEW DIRECTLY ON THE SITE.





- **FINALLY, THE GENERATED TOOLPATHS WILL BE VISUALIZED IN 3D – BOTH CONTOUR TRAJECTORY AND SPIRAL TRAJECTORY. THE USER WILL ALSO BE ABLE TO DOWNLOAD A ZIPPED FILE CONTAINING THE CORRESPONDING TOOLPATH POINTS BY CLICKING ON THE RESPECTIVE CONTOUR OR SPIRAL DOWNLOAD BUTTONS.**

